

AHMED ELBAKLY

Senior Backend Engineer | Node.js · NestJS · TypeScript · Distributed & Multi-Tenant SaaS Systems

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years architecting and operating production SaaS platforms on Node.js, NestJS, Express.js and TypeScript. Delivered multi-tenant backends spanning 90+ feature modules, secured by JWT authentication, RBAC and audit logging, and scaled through Redis caching, BullMQ queues, cron-driven jobs and Socket.IO real-time services over MongoDB and PostgreSQL. Owns the full lifecycle — system design, REST API architecture, microservices patterns, Dockerised delivery, GitLab CI/CD and cloud operations on AWS, Azure AKS and DigitalOcean. Sole technical owner on 9+ client platforms across real estate, EdTech, e-commerce and on-demand services. Bilingual (Arabic/English) engineer serving MENA, European and remote US teams.

CORE COMPETENCIES

Backend Architecture · System Design · REST API Design · Microservices Patterns · Event-Driven & Queue-Based Processing · Multi-Tenant SaaS Architecture · Authentication & Authorization · JWT · RBAC · API Security · Redis Caching · Performance Optimization · Database Modeling & Query Tuning · Real-Time Systems (WebSockets) · Background Jobs & Scheduling · Cloud Deployment · CI/CD Automation · Containerization & Orchestration · Third-Party & Payment Integrations · Technical Ownership & Client Delivery

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL, HTML5, CSS3

Backend: Node.js, NestJS, Express.js, REST APIs, GraphQL, Microservices Patterns, Multi-Tenant Architecture, Socket.IO / WebSockets, BullMQ, Cron Jobs, JWT, RBAC

Databases & Caching: MongoDB (Mongoose), PostgreSQL, MySQL, Redis, Prisma

Cloud: AWS (EC2, S3), Azure AKS, DigitalOcean, Cloudflare (DNS / SSL / CDN)

DevOps: Docker, Kubernetes, GitLab CI/CD, Nginx, Linux (Ubuntu)

Testing: Jest, Postman

Frontend: React 18/19, Next.js (App Router), Redux Toolkit, RTK Query, TanStack Query, Material UI, Tailwind CSS

Integrations: Stripe (billing & subscriptions), Keycloak (OIDC / OAuth2), Firebase, Twilio, Agora, LiveKit, Google Maps, Mapbox

Tools & Practices: Git, GitHub, GitLab, Agile / Scrum, Code Review, i18n & RTL (EN-AR)

PROFESSIONAL EXPERIENCE

Senior Backend Engineer — Freelance / Contract May 2020 – Present

Remote — SaaS, CRM, EdTech, E-Commerce & On-Demand Platforms

- Architected multi-tenant backend platforms in NestJS and Node.js for 9+ production clients, isolating tenant data and configuration on a shared codebase so new customers could be onboarded without forking the product.
- Engineered secure REST APIs with JWT authentication, role-based access control and audit logging, enforcing least-privilege access across six distinct user roles and creating a traceable record for compliance-sensitive clients.
- Scaled asynchronous workloads with BullMQ queues and scheduled cron jobs, moving notifications, reporting and recurring operations off the request path to keep API latency predictable under load.
- Optimized data access through Redis caching and refined MongoDB and PostgreSQL query design, cutting repeated database reads and improving response times on the highest-traffic endpoints.
- Designed real-time services — chat, push notifications and live order tracking — on Socket.IO and WebSockets, replacing polling with event-driven delivery across several production systems.
- Integrated Stripe billing and subscriptions, Keycloak identity, Firebase, Twilio SMS, Agora and LiveKit streaming, Google Maps and Mapbox, extending product capability without rebuilding core services.
- Automated releases by containerizing services with Docker and building GitLab CI/CD pipelines, replacing manual deployment steps with repeatable, reviewable builds across environments.
- Operated production workloads on AWS (EC2, S3), Azure AKS (Kubernetes) and DigitalOcean, fronted by Nginx reverse proxies and Cloudflare DNS, SSL and CDN for secure, resilient traffic handling.
- Led delivery end to end as the sole technical owner — requirements, system design, estimation, code review and production troubleshooting — for clients with no in-house engineering function.

KEY PROJECTS

49Hub — Multi-Service Super-App Backend Engineer

Node.js · TypeScript · Express.js · MongoDB · Redis · Socket.IO · BullMQ · AWS S3

- Consolidated ride-hailing, food delivery, hotel booking, live streaming, social networking and marketplace services into a single backend of 90+ feature modules, avoiding a fragmented estate of disconnected apps.
- Structured the service layer around shared domain models and modular boundaries, so each vertical could evolve independently while reusing authentication, payments and notification infrastructure.
- Strengthened throughput with Redis caching for hot reads, BullMQ workers for heavy background processing and Socket.IO for live location and messaging flows.
- Secured multi-role access with RBAC and audit logging, and integrated Agora, LiveKit, Google Maps, Mapbox, Firebase, AWS S3 and multiple payment gateways.

Schoolie — Multi-Tenant School Management SaaS Full-Stack Engineer

NestJS · React 18 · TypeScript · MongoDB · Redux Toolkit · RTK Query · AWS S3 · Firebase · Twilio · Jest

- Replaced fragmented school administration workflows with a single multi-tenant SaaS platform of 43 backend modules covering academics, attendance, payroll, fee collection, communication and reporting.
- Modeled granular RBAC across Admin, School Manager, Teacher, Student, Parent and Driver roles, keeping sensitive academic and payroll data visible only to the right audience per tenant.
- Automated recurring school operations with cron jobs and integrated Twilio SMS, Firebase notifications, AWS S3 storage and Excel export pipelines, removing manual administrative effort.
- Covered core services with Jest tests and delivered real-time chat, analytics dashboards and full English/Arabic support to widen adoption across MENA schools.

Coonex — Real Estate CRM SaaS Full-Stack Engineer

Node.js · Express.js · MongoDB · Next.js 13 · React 18 · TypeScript · CASL · Socket.IO · Stripe

- Delivered a multi-tenant CRM that gave real estate agencies one system for 500+ properties and 1,000+ leads, superseding spreadsheet-based lead and pipeline tracking.
- Modernized monetization by integrating Stripe billing and subscription management, enabling recurring revenue and self-service plan changes for agency customers.
- Enforced permissioning with CASL-based authorization and delivered instant updates through Socket.IO notifications across sales pipelines, meetings and documents.
- Shipped a separate B2B admin dashboard with RBAC, analytics, reporting and invoice management, giving the operator central control over tenants and billing.

Additional Delivery Backend & Frontend

- **BaladiCRM** — CRM platform with 10+ business modules, an AI chatbot flow builder with training and approval workflows, and a scalable Redux architecture (21+ slices) secured by JWT and RBAC. (React 18, Redux Toolkit, Socket.IO, MUI, Vite)
- **Jumla** — Multi-vendor e-commerce and delivery dashboard: real-time order tracking, RBAC workflows and Google Maps geolocation, deployed and maintained on Docker, Kubernetes and Azure AKS.
- **Luminar** — Feature-based e-commerce admin dashboard (products, orders, payments, revenue reporting, audit logs) plus a multilingual EN/AR corporate site on the Next.js App Router. (React 19, TypeScript, TanStack Query)

CLOUD & DEVOPS EXPERIENCE

- Deployed and maintained production applications across AWS, Azure AKS, DigitalOcean and Hostinger environments, handling server provisioning, monitoring and live incident troubleshooting.
- Built GitLab CI/CD pipelines for automated testing, deployment and release, and containerized services with Docker for consistent behaviour from development to production.
- Configured Nginx reverse proxies and managed domains, DNS, SSL certificates and CDN delivery through Cloudflare to improve availability and transport security.

EDUCATION & TRAINING

- **Bachelor of Commerce** — Nile Academy for Science, Egypt (2012)
- **Web Development — Advanced, Professional & Challenger Tracks** — Udacity (FWD Initiative)

LANGUAGES

Arabic — Native | English — Professional Working Proficiency